

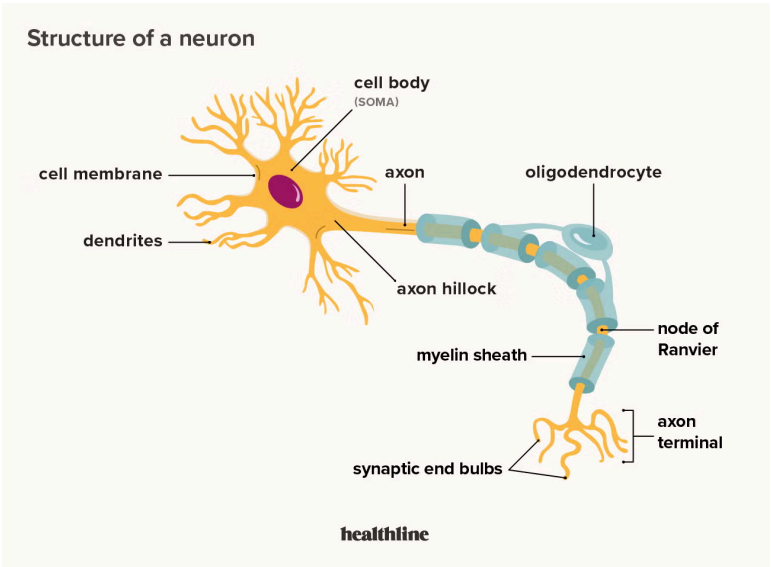
Human Decision Making

🕒 Created time	@April 19, 2025 1:35 PM
🎯 Goal Relation	🧠 Launch AlphaPi
🕒 Last edited time	@April 19, 2025 2:00 PM

Appendix : Data collection

Important Concepts

Neurons



Structure	Where it is	Main job	What to remember
Dendrites	Short, branching "antennae" that sprout from the cell body	Receive chemical signals from other neurons and funnel the electrical	A neuron can have dozens to <i>thousands</i> of dendritic branches, giving it a big "listening surface."

		changes toward the soma	
Soma / cell body	Central bulb that contains the nucleus, mitochondria, ribosomes, etc.	Integrates all incoming dendritic signals; if voltage crosses threshold the soma triggers an action potential in the axon	Think of it as the neuron's command center and metabolic engine.
Axon (with myelin and nodes of Ranvier)	Long cable that extends from the soma, often centimetres long	Carries the all-or-nothing electrical impulse (action potential) away from the soma toward other cells	Myelin "insulation" lets the signal jump node-to-node, making conduction up to 100× faster.
Axon terminals (a.k.a. terminal boutons, synaptic end bulbs)	Tiny swellings at the far end of each axon branch	Convert the electrical impulse into chemical form—releasing neurotransmitter across a synapse to the next cell	One axon can split into hundreds of terminals, broadcasting the message widely.
Oligodendrocyte (glial cell, not part of the neuron)	Star-shaped cell whose many arms wrap around CNS axons	Produces the myelin sheath; one oligodendrocyte can insulate segments on dozens of neighbouring axons	The peripheral-nervous-system equivalent is the <i>Schwann cell</i> . Damage to oligodendrocytes leads to demyelinating diseases like MS.

Types of decision Making

Got from ChatGPT as a summary

Rank → Fastest ► Slowest	System & "job"	Typical latency / integration window*
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0. Monosynaptic spinal reflexes Postural & withdrawal "no-brainers"	20–50 ms (M1 burst $\approx$ 44 ms)	One sensory synapse onto α -motor neurons in the spinal cord
1. Trans-cortical & brain-stem sensorimotor loops Fast corrective movements, startle, VOR	70–100 ms (long-latency M2 $\approx$ 69 ms; M3 $\approx$ 95 ms)	Spinal cord $\leftrightarrow$ cerebellum $\leftrightarrow$ primary motor cortex; tiny "micro-plans" adjust the reflex before the cortex can "think."
2. Rapid affective appraisal ("low-road" emotion) Detect threat or salience	120–250 ms to first spikes in the amygdala for faces, snakes, etc.	Superior colliculus & pulvinar deliver coarse sensory gist straight to the amygdala, priming autonomic and orienting responses.
3. Habit / procedural controller Lean on well-learned stimulus- $\rightarrow$ -response chains ("autopilot")	200–500 ms ; BG neurons show longer intrinsic timescales than downstream motor nuclei	Loops through basal ganglia, cerebellum & premotor cortex gate or suppress the next action without full cortical deliberation.
4. Perceptual evidence accumulator ("System 1" intuition) Choose between competing options as evidence builds	0.3–1 s for routine two-choice tasks; centro-parietal positivity & LIP ramps emerge $\approx$ 300 ms after stimulus	Parietal cortex (LIP/IPS), premotor & frontal eye field implement a drift-diffusion-like race to bound.
5. Working-memory & rule-guided decisions Hold several items/rules, re-evaluate, inhibit impulses	1–10 s integration windows; PFC oscillations maintain information over seconds	Dorsolateral & medial prefrontal cortex, anterior cingulate, fronto-hippocampal theta-gamma coupling keep task goals online.
6. Multi-step analytical reasoning ("System 2") Plan a route, solve an algebra proof, weigh pros/cons	10 s–minutes ; hierarchical PFC & hippocampal replay string together sub-decisions	Anterior PFC, dorsomedial PFC, hippocampus and parietal cortex pass partial results back and forth; activity spans sentences, chess moves, code blocks.
7. Extended deliberation & mind-wandering Creative synthesis, life planning, narrative thinking	Minutes–hours of integration; highest-order association areas track	Default-mode network (medial PFC, posterior cingulate, angular gyrus) with slow intrinsic timescales pools

	storylines over many minutes	information across whole episodes.
8. Neuromodulatory & hormonal regulators Bias the whole ladder toward vigilance, hunger, sleep, etc.	Hours – days	Hypothalamus, brain-stem neuromodulators (dopamine, norepinephrine, serotonin) and endocrine signals tune gain and thresholds for every faster layer underneath.